OPENSOURCESDRLAB DOCUMENTATION

PlutoSky R2 Quick Start Guide

Product	PlutoSky R2
Document	OSSDRLAB-PlutoSky-R2-QSG
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1. Overview

The PlutoSky R2 is a software-defined radio based on the Analog Devices ADALM-PLUTO design, with a Zynq XC7Z020-2CLG484I, 1 GB DDR3, an AD9361 or AD9363 transceiver (2R2T), Gigabit Ethernet on the FPGA and a 10 MHz reference input. It works with PlutoSDR software over USB or Ethernet.

Two firmware sets are installed at the factory: the QSPI flash holds a port of the Analog Devices PlutoSDR firmware v0.38, and the microSD card holds the Tezuka firmware. This guide covers the first connection, the boot modes and re-installing the firmware. The full user manual is on the blog: https://blog.opensourcesdrlab.com/archives/plutosky-r2-user-manual.

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2. What you need

Item	Notes
PlutoSky R2	
USB-C cables	one for the **USB2.0** port (data), one for the **DEBUG** port (serial console, JTAG)
PC	Windows, Linux or macOS
PlutoSDR USB drivers (Windows), libiio, IIO Oscilloscope	from Analog Devices: https://wiki.analog.com/university/tools/pluto/users
Serial terminal	for example PuTTY, 115200 baud
Firmware files (only to re-install)	from the Releases page
Vivado/Vitis or Vivado Lab (only to write the QSPI flash over JTAG)	from AMD: https://www.xilinx.com/support/download.html

3. Board connectors

Connector	Function
TX1, RX1, TX2, RX2	SMA RF ports
10M_IN	10 MHz reference input; the LOCK LED shows that the 40 MHz oscillator is locked to it
ETH	Gigabit Ethernet (connected to the FPGA)
DEBUG (USB-C, middle)	JTAG and UART console (FT2232H)
USB2.0 (USB-C)	USB network adapter, USB drive and serial console of the PlutoSDR firmware; power
TF	microSD card slot
BOOT	2-position boot switch (see section 4)
RST	reset button
GPIO header	expansion I/O (1.8 V)
DONE, LED, PWR	FPGA configured, status, power

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4. Boot modes

Mode	Switch 1	Switch 2	Boots from
QSPI (factory setting)	1 (up)	0 (down)	on-board QSPI flash (PlutoSDR firmware v0.38 port)
SD	1	1	microSD card (Tezuka firmware as shipped)
JTAG	0	0	nothing; for programming and debugging over JTAG

Change the switch only while the board is powered off.

5. First connection

1. Windows: install the PlutoSDR USB drivers from Analog Devices, then libiio.
2. Connect the **USB2.0** port. The board appears as a network adapter (board `192.168.2.1`), a USB drive and a serial port.
3. Check: `iio_info -u ip:192.168.2.1`, then connect IIO Oscilloscope, GNU Radio or another PlutoSDR program to `ip:192.168.2.1`.
4. Ethernet: the v0.38 port uses `192.168.1.10` by default. To change it, log in on the DEBUG serial console (user `root`, password `analog`) and run `fw_setenv ipaddr_eth <address>`, then reboot.

> **Note:** All boards running the v0.38 port report the same USB serial number. To use several boards on one PC, give each board its own IP address (`config.txt` on the board's USB drive) and connect by IP.

6. Re-installing the firmware

SD card (PlutoSDR firmware v0.38 port)

1. Format a microSD card as FAT32 and copy all files from `PlutoSky-R2_SD-card-firmware_v0.38.zip` to it.
2. Set the boot switch to SD, insert the card and power the board.

For the Tezuka firmware, follow https://blog.opensourcesdrlab.com/archives/plutosky-r2-tezuka-firmware-user-manual.

QSPI flash (factory firmware)

1. Unzip `PlutoSky-R2_QSPI-flash-firmware_v0.38.zip`.

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2. Set the boot switch to JTAG, connect the **DEBUG** port and write `boot.bin` to the QSPI flash with Vivado/Vitis *Program Flash* (flash type `qspi-x4-single`, FSBL `fsbl.elf`).
3. Power off, set the boot switch to QSPI and connect the **USB2.0** port: the board starts in DFU mode.
4. Run `program_dfu.bat` (Windows; uses `dfu-util` from the PlutoSDR USB drivers). On Linux run the two commands in the script with `dfu-util` directly.
5. Power-cycle the board.

openwifi

`jtag_spi_openwifi.zip` writes the openwifi image to the QSPI flash over JTAG. Follow its release notes and https://blog.opensourcesdrlab.com/archives/plutosky-r2-openwifi-user-manual. Login: user `root`, password `openwifi`.

7. Files in this repository

Path	Contents
`hardware/PlutoSky-R2_Schematic.pdf`	Board schematic
`firmware/source/`	Source changes of the PlutoSDR firmware port against Analog Devices plutosdr-fw v0.38, with build steps

8. Support

- Product page: https://opensourcesdrlab.com/products/plutosky-r2
- User manual: https://blog.opensourcesdrlab.com/archives/plutosky-r2-user-manual
- Email: support@opensourcesdrlab.com
- Forum: https://forum.opensourcesdrlab.com
- Discord: https://discord.gg/nnxY6DkmvT

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